

MedTrack_DV — Healthcare Operations Methodology

Project: MedTrack_DV — Hospital Operations & Patient Analytics Dashboard (Group 2) **Mentor:** Pathina Likhita

1. Introduction

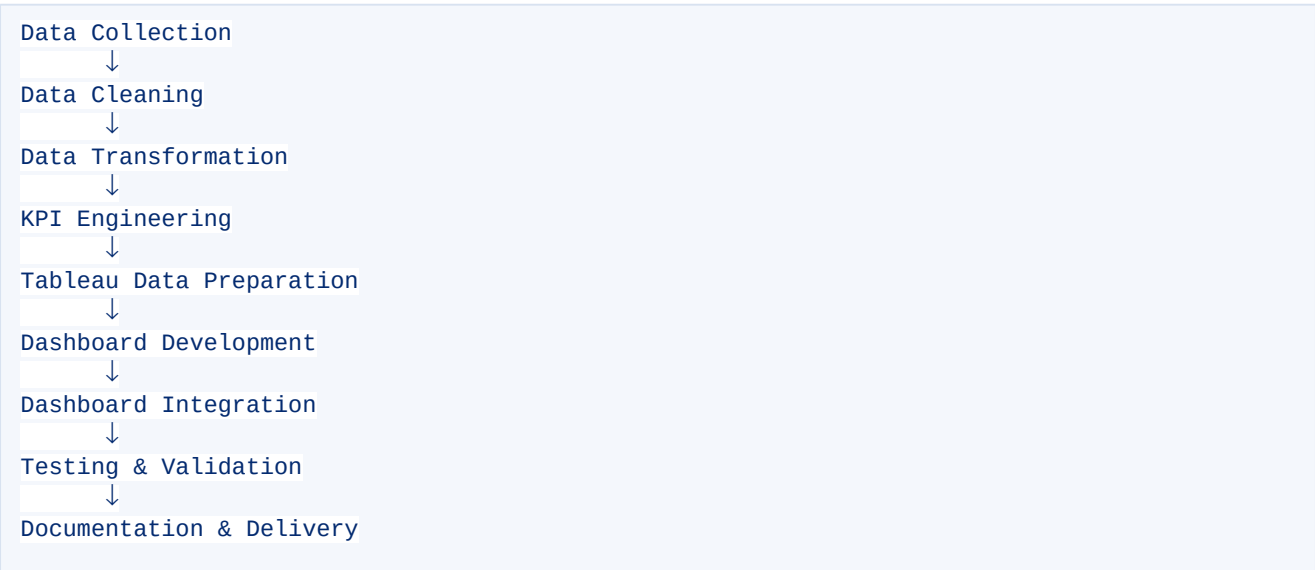
MedTrack_DV is a healthcare operations analytics project developed using data preparation and Tableau visualization.

The project transforms patient and hospital operational data into meaningful analytical dashboards.

The analysis focuses on:

- Hospital performance
- Patient flow
- Department performance
- Resource utilization

2. End-to-End Methodology



3. Stage Details

#	Stage	Focus	Key Output
1	Data Collection	Download hospital operational datasets; collect patient admission records; gather department and resource data; integrate datasets	Integrated healthcare dataset
2	Data Cleaning	Remove duplicates; handle missing patient data; validate record counts	Consistent record set
3	Data Transformation	Standardize department names; normalize healthcare indicators; derive categories (age group, wait-time category, shift)	Cleaned & enriched dataset
4	KPI Engineering	Engineer the 12 operational KPIs (patients, admissions, LOS, occupancy, readmission, utilization, wait time, satisfaction, efficiency)	KPI-ready dataset
5	Tableau Data Preparation	Tableau-ready CSV; typed fields; calculated fields (Total Admissions , Average LOS)	Tableau data source
6	Dashboard Development	Build the four dashboards worksheet by worksheet	MedTrack_DV workbook
7	Dashboard Integration	Global filters, navigation controls, parameter actions, dashboard linking	Integrated suite
8	Testing & Validation	Validate KPI calculations (>95% accuracy); verify healthcare metrics; test interactions	QA checklist & test report
9	Documentation & Delivery	Dataset sources, KPI definitions, dashboard guide, methodology; organized repository; GitHub delivery	Final documentation package

4. Milestones

Milestone	Weeks	Modules	Focus
Milestone 1	Weeks 1–2	Module 1, Module 2	Data Collection & Cleaning
Milestone 2	Weeks 3–4	Module 3, Module 4	KPI Engineering & Dashboard Planning
Milestone 3	Weeks 5–6	Module 5, Module 6	Dashboard Development & Integration
Milestone 4	Weeks 7–8	Module 7, Module 8	Testing and Delivery

5. Tools Used per Stage

Stage	Tools
Data Collection	Python
Cleaning & Transformation	Python (Pandas, NumPy)
KPI Engineering	Python (Pandas, NumPy)
Dashboard Development	Tableau Desktop
Dashboard Integration	Tableau Filters, Parameters, Actions
Testing	Manual QA + Python cross-checks
Documentation	Markdown, GitHub

6. Future Scope

The methodology is designed to extend beyond the delivered project:

1. **Real-time data pipeline** — stream live admissions into the dashboards instead of batch refreshes.
2. **Predictive analytics** — forecast admissions, occupancy and length of stay; predict readmission risk.

3. **Automated alerting** — notify stakeholders when occupancy crosses the 85% threshold or waits exceed targets.
4. **Bed- and ward-level drilldown** — from hospital KPIs to individual bed status.
5. **Multi-hospital benchmarking** — compare departments across hospitals and regions on one KPI framework.
6. **Natural-language Q&A** — ask the dashboard questions in plain language.

These enhancements would evolve MedTrack_DV from an analytical reporting suite into a real-time operational decision platform.

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